



# Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy

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## INTRODUCTION

Genitourinary syndrome of menopause (GSM; previously referred to as vaginal atrophy, vulvovaginal atrophy, urogenital atrophy, or atrophic vaginitis) is defined as a collection of symptoms and signs caused by hypoestrogenic changes to the labia majora/minora, clitoris, vestibule/introitus, vagina, urethra, and bladder that occur in perimenopausal and menopausal patients. The syndrome may include, but is not limited to, genital symptoms of dryness, burning, and irritation; sexual symptoms of lack of lubrication, discomfort or pain, and impaired function; and urinary symptoms of urgency, dysuria, and recurrent urinary tract infections.

The spectrum of adverse consequences makes long-term treatment with hormonal and nonhormonal interventions essential in many patients, not only for relief of symptoms but also improved quality of life [1,2]. Treatment options include both hormonal and nonhormonal interventions.

This topic discusses the management of vulvovaginal symptoms related to GSM. The evaluation of GSM, the treatment of other GSM symptoms, including urinary symptoms and sexual pain, and the treatment of other symptoms related to menopause are presented separately.

- (See "[Menopause: Clinical features and diagnosis](#)".)
- (See "[Female urinary incontinence: Treatment](#)".)

- (See ["Recurrent simple cystitis in female adults".](#))
  - (See ["Overview of sexual dysfunction in females: Management".](#))
  - (See ["Treatment of menopausal symptoms with hormone therapy".](#))
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## CLINICAL APPROACH

The primary indication for treatment of GSM is to relieve symptoms that cause distress in a patient who has diminished ovarian estrogen production due to menopause or other etiologies. Vulvovaginal symptoms include vaginal dryness, burning, pruritus, dyspareunia, vaginal discharge, bleeding, or spotting. Urinary tract symptoms include dysuria, urinary incontinence, urinary frequency, urethral discomfort, or, infrequently, hematuria. (See ["Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms", section on 'Clinical presentation'.](#))

- **Treatment barriers** – Many patients with GSM do not seek medical care for these issues (due to embarrassment, cultural, religious, or societal beliefs, or lack of awareness about safe and effective treatments) and many clinicians do not ask their menopausal patients about GSM symptoms [3]. This is discussed in detail separately. (See ["Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms", section on 'Clinical presentation'.](#))

Barriers to effective management include product cost and labeling. High cost of a specific therapy may lead a clinician to recommend a less expensive (and less effective) over-the-counter alternative which may improve mild GSM symptoms, but does not treat the underlying pathology (ie, decline in serum estrogen levels). Issues related to product labeling (as required for all estrogen-containing therapies used to treat menopausal symptoms) are discussed in detail below. (See ["Vaginal estrogen therapy"](#) below.)

- **Step-wise approach** – Management is guided by the patient's symptoms, overall medical health, and goals. For symptoms of vaginal dryness, discomfort, or dyspareunia associated with vaginal atrophy, first-line treatments are often nonhormonal vaginal moisturizers and lubricants [4]. These are particularly helpful for those with mild symptoms or may be the best alternative for those with estrogen-sensitive neoplasias (see ["Patients with breast cancer"](#) below). If first-line therapies do not provide adequate symptom relief, vaginal estrogen therapy or other hormonal medications may be prescribed for patients without contraindications to their use [5]. Vaginal estrogen therapy may also help to ameliorate some urinary tract symptoms of GSM, including frequent urinary tract infections or urinary incontinence in the absence of infection [6]. This is discussed in more detail separately. (See ["Female urinary incontinence: Treatment", section on 'Topical vaginal estrogen'.](#))

Most patients with GSM achieve adequate symptom relief with vaginal hormone therapy, but other modes of treatment, such as physical therapy or vaginal dilators, may be used for refractory symptoms or if hormone therapy is contraindicated.

- **Conditions to exclude** – Prior to initiating treatment for vaginal atrophy, other conditions should be excluded, in particular:
  - Patients with postmenopausal bleeding should be evaluated for endometrial hyperplasia, endometrial polyps, or cancer. (See ["Overview of the evaluation of the endometrium for malignant or premalignant disease"](#).)
  - Patients with postcoital bleeding should be evaluated for infectious, noninfectious (eg, cervical ectropion, cervical polyp), and premalignant or malignant conditions. (See ["Postcoital bleeding in females"](#).)
  - Patients with urinary tract symptoms should be evaluated for urinary tract infection or other conditions (interstitial cystitis, urinary tract malignancy), as appropriate. (See ["Acute simple cystitis in female adults"](#) and ["Interstitial cystitis/bladder pain syndrome: Clinical features and diagnosis"](#).)
  - Patients with symptoms of vaginitis should be evaluated for infectious and noninfectious etiologies. (See ["Vaginitis in adults and adolescents: Initial evaluation"](#).)
  - Patients with a strong family history of breast cancer should be referred for genetic testing before treatment. (See ["Genetic testing for hereditary breast and ovarian cancer syndromes"](#).)

A detailed discussion regarding the differential diagnosis of symptomatic urogenital atrophy is found elsewhere. (See ["Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms"](#), section on 'Differential diagnosis'.)

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## INITIAL THERAPY WITH MOISTURIZERS AND LUBRICANTS

First-line therapy for GSM includes nonhormonal vaginal moisturizers and lubricants. Symptoms of vaginal dryness can be managed by the regular use of vaginal moisturizing agents with supplemental use of vaginal lubricants for sexual intercourse [2]. These agents may improve coital comfort and increase vaginal moisture, but they do not reverse most atrophic vaginal changes [4]. Thus, while vaginal moisturizers and lubricants are useful for patients with mild symptoms, many patients will have persistent symptoms and require hormonal medications or other treatments [7].

**Product types and use** — Patients should be counseled about the difference between vaginal moisturizers and lubricants and how to use them. Patients should try various products until they find one that meets their needs and to find the one they like best and within their price range. The only contraindication is an allergy to a particular product.

- **Vaginal moisturizers** are intended for use routinely, typically two or three days per week, not just during sexual activity. These products are typically bioadhesives. Hyaluronic acid is often used as a key ingredient in vaginal moisturizers. Many moisturizer products are available in pharmacies and online (eg, Replens, Vagisil Moisturizer, Feminase, Moist Again, K-Y Liquibeads, Hyalo GYN, Revaree suppositories) ( [table 1](#)).

- **Lubricants** are used only at the time of sexual activity, both with and without vaginal penetration ( [table 2](#)). Lubricants are classified as:

- Water-based – Examples include Astroglide, Slippery Stuff, K-Y Jelly, and Goodlove. Some clinicians counsel their patients that water-based lubricants with [glycerin](#) (also referred to as glycerol) may result in an increased risk of vulvovaginal candidiasis, but there is no evidence to support this, and some data suggest that glycerin inhibits candidal growth [8,9].
- Silicone-based – Examples include Uberlube, Swiss Navy Premium Silicone, Astroglide X LiquiGel, Jo Premium (Silicone line), Valm Silicone Lube, Pjur, and ID Millennium. Such lubricants may be better for extended use or water-based play. They also tend not to dry out or alter pH levels and are generally hypoallergenic. While they are compatible with condoms, they can degrade silicone sex toys.
- Oil-based lubricants – Examples include Elegance Women's Lubricant, Simply Slick, and AH! Yes OB. These often use ingredients like coconut oil, shea butter, or cocoa butter, which are not compatible with latex condoms as they can cause breakdown.

Commonly available nonhormonal vaginal moisturizers formulations

| Product             | Ingredients                                                                   | pH 1-7 | Osmolality (mOsm/kg) | Recommendations and use                       | Approximate price |
|---------------------|-------------------------------------------------------------------------------|--------|----------------------|-----------------------------------------------|-------------------|
| Replens             | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| Vagisil Moisturizer | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| Feminase            | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| Moist Again         | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| K-Y Liquibeads      | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| Hyalo GYN           | Hyaluronic acid, polyethylene glycol, propylene glycol, glycerin, water       | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |
| Revaree             | Hydroxyethylcellulose, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply 2-3 times per week or after intercourse | \$14.99 (3.33 oz) |

Commonly available nonhormonal vaginal moisturizers formulations

Table 1 - larger image below

Commonly available nonhormonal lubricants

| Product                     | Ingredients                                                   | pH 1-7 | Osmolality (mOsm/kg) | Recommendations and use                | Approximate price |
|-----------------------------|---------------------------------------------------------------|--------|----------------------|----------------------------------------|-------------------|
| Astroglide                  | Water, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Slippery Stuff              | Water, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| K-Y Jelly                   | Water, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Goodlove                    | Water, polyethylene glycol, propylene glycol, glycerin, water | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Uberlube                    | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Swiss Navy Premium Silicone | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Astroglide X LiquiGel       | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Jo Premium (Silicone line)  | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Valm Silicone Lube          | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| Pjur                        | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |
| ID Millennium               | Silicone oil, dimethylsiloxane, water, glycerin, water        | ~4.5   | ~280-300             | Apply as needed during sexual activity | \$14.99 (3.33 oz) |

Commonly available nonhormonal lubricants

Table 2 - larger image below

Both moisturizers and lubricants are also available in preparations with fewer preservatives (eg, Astroglide [Glycerin](#) & Paraben Free, Sliquid Naturals H2O, ID Free Hypoallergenic).

**Efficacy** — Vaginal moisturizer and lubricant therapy appear to be effective in many individuals, and some data suggest that they are as effective as vaginal estrogen [10]. However, data are limited by sample size and duration of follow-up; more study is needed of this issue.

The largest randomized trial regarding treatment options for GSM found few differences in efficacy between vaginal estrogen and two nonhormonal lubricants/moisturizers but had significant methodological limitations [5]. The trial assigned 302 postmenopausal patients with vulvovaginal symptoms to one of three vaginal therapy regimens: [estradiol](#) tablet (10 mcg) and placebo gel, vaginal moisturizer (Replens) and placebo tablet, or placebo tablet and placebo gel (dual placebo). The vaginal moisturizer and placebo gel had similar ingredients, but the placebo did not contain the bioadhesive polymer [polycarbophil](#), which is intended to improve efficacy.

- The primary outcome measure was decrease in severity of most bothersome symptom (eg, vulvovaginal pain, itching, dryness). At 12 weeks, all groups had similar reductions in severity from a baseline score of approximately 2.5 (moderate to severe) to 1.3 (mild to moderate); no significant differences in severity of most bothersome symptom were found between groups. The only significant differences found were in scores on the Vaginal Symptom Index, with dual placebo compared with moisturizer found to be slightly superior in improvement in pain with vaginal penetration. Improvement in mean Female Sexual Function Index scores was also similar across groups.
- A separate report using data from the trial found greater improvement in menopause-specific quality of life scores in the [estradiol](#) tablet group compared with the other two groups [11].
- In a subsequent post hoc analysis that assessed the impact of 12 weeks of treatment, the patient-reported number of days with sexual activity in the past month, sexual activity in the past week, and penetrative sex in the past week, as well as average pain with sexual activity, were similar for both interventions compared with placebo [12]. Major limitations of this post hoc analysis include that it only had the power to detect large differences in outcomes and had insufficient power to allow direct comparison of the active treatments (vaginal estrogen versus moisturizer).

A major limitation of this study design overall was that it is uncertain whether it is generalizable to clinical care since the standard of care is to use nonhormonal interventions first and to limit vaginal estrogen therapy to patients who fail nonhormonal management. Also, the study regimens were not consistent with usual therapy (all patients had five days per week of treatment, while the majority of patients likely use vaginal estrogen or a moisturizer

two to three days per week). Limitations in study design were that all groups were treated with two interventions concurrently, so it did not provide a direct comparison between the three interventions. Also, the placebo gel was similar to commonly used products and thus was not a nonactive comparator, although this would be difficult to avoid.

Other data regarding the efficacy of vaginal moisturizers include a small randomized trial and a small observational study that compared a vaginal moisturizer (Replens; contains multiple ingredients, including [mineral oil](#)) and a vaginal estrogen preparation and reported that relief of vaginal dryness and restoration of normal vaginal pH were similar for both agents [13,14]. In contrast with estrogen, Replens does not affect the vaginal maturation index [15]. In our experience, however, estrogen is more effective than vaginal moisturizers.

There are inconsistent data regarding whether moisturizers or lubricants that result in an acidic pH (in premenopausal patients, normal vaginal secretions have a pH of 4.0 to 4.5) are more effective. As an example, a randomized trial of postmenopausal breast cancer survivors with vaginal atrophy found that a gel with added lactic acid (pH 4.0) decreased vaginal irritation, dryness, and dyspareunia compared with a gel without lactic acid (pH 7.2) [16]. However, a subsequent trial found no improvement in dyspareunia with a pH-balanced gel compared with placebo [17]. Further study is needed to validate these results. (See "[Vaginitis in adults and adolescents: Initial evaluation](#)", section on 'Normal vaginal discharge'.)

**Role of pelvic floor muscle exercises** — First-line therapy may also include pelvic floor muscle (Kegel) exercises. Although these can usually be taught to the patient during the clinic visit, referral for pelvic floor therapy training can be helpful. This intervention is discussed in detail separately. (See "[Patient education: Pelvic floor muscle exercises \(Beyond the Basics\)](#)".)

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## PERSISTENT SYMPTOMS

Vaginal estrogen therapy is a common second-line treatment for GSM that does not adequately respond to moisturizers and lubricants.

Hormonal medications (eg, vaginal dehydroepiandrosterone [DHEA], oral [ospemifene](#)) other than estrogen are also available for GSM treatment; [testosterone](#) may be used off-label in select patients. These are less commonly used, but patients should be counseled about all treatment options, and some may prefer another medication. There are also other types of treatment available, including complementary medications or laser surgery, but the safety and efficacy of these are uncertain.

**Vaginal estrogen therapy** — Low-dose vaginal estrogen is the most effective treatment for moderate to severe symptoms of vaginal atrophy not responsive to nonhormonal intervention. Use of vaginal estrogen therapy is appropriate for patients with symptoms of vaginal atrophy in the setting of low estrogen levels, provided that there are no

contraindications to this therapy (eg, some patients with estrogen-dependent tumors, particularly those on antiestrogen therapy).

Adequate vaginal estrogen therapy leads to restoration of the normal vaginal acidic pH and microflora, thickening of the epithelium, increased vaginal secretions, and decreased vaginal dryness and resultant dyspareunia. In addition, vaginal estrogen therapy is associated with urinary tract benefits. These include a reduction in the incidence of urinary tract infections and overactive bladder symptoms. Urinary incontinence, however, typically requires other modes of treatment [6]. (See ["Recurrent simple cystitis in female adults"](#), section on 'Topical estrogen for postmenopausal females' and ["Female urinary incontinence: Treatment"](#).)

Vaginal estrogen therapy is also likely to effectively treat vaginal discharge if the sole etiology is hypoestrogenism. If vaginal discharge continues despite vaginal estrogen therapy, the patient should be evaluated for other etiologies of vaginitis. Of note, vaginal estrogen creams will leak out of the vagina for a few days after application and may appear to be discharge; however, the creamy consistency of this leakage typically makes it easy to differentiate from pathologic vaginal discharge. (See ["Vaginitis in adults and adolescents: Initial evaluation"](#).)

Adverse effects of vaginal estrogen therapy are uncommon [18]. Patients may complain of vaginal irritation or itching, vaginal bleeding, or breast tenderness; headaches or hot flashes are rare.

**Serum absorption** — Vaginal estrogen therapy at recommended doses results in estrogen absorption into the circulation, although to a lesser degree than oral or transdermal estrogen therapy. The serum [estradiol](#) levels attained with the low-dose preparations are slightly higher than the average level for postmenopausal patients, which is approximately 5 pg/mL [19,20]. For the same dose of estrogen, a vaginal dose results in lower serum estradiol and [estrone](#) levels than oral estrogen (30 percent lower in a study of [conjugated estrogens](#)) [21].

In general, the plasma [estradiol](#) assays used in studies of vaginal estrogen absorption have not been sufficiently sensitive to measure the low serum levels of estradiol in menopausal patients and to measure compounds other than estradiol [19]. Measurements of estradiol by mass spectrometry assays or ultrasensitive bioassays find much lower levels in postmenopausal patients than do radioimmunoassays [22]. When highly sensitive assays were used, increments in estradiol can be detected at very low levels (ie, 5 pg/mL) but not to premenopausal levels (ie, 40 to 600 pg/mL) [23].

The effect of the severity of vaginal atrophy on systemic estrogen absorption is controversial. Some studies have reported that systemic absorption is highest in the first days or weeks of therapy and then decreases with ongoing treatment [24,25]; however, a study using highly sensitive [estradiol](#) measurements did not support this finding [23]. When present, a higher level of absorption with initial therapy may be due to increased estrogen

absorption through the thin atrophic vaginal epithelium, which then cornifies with treatment [24]. In patients who use the low-dose vaginal ring, another possibility is that the device may release a higher dose of estradiol during the first 24 hours of use. This was illustrated in an in vitro study in which the ring released up to 55 mcg during day 1 of testing, versus 7 to 8 mcg during days 2 to 90 [26].

There are no long-term randomized trials evaluating the safety of low-dose vaginal estrogen use. However, long-term observational data (with a median duration of use up to three years) have not shown any increased risk of breast or endometrial cancer, coronary artery disease, stroke, or venous thromboembolism [27-30]. In addition, the safety of vaginal estrogen on the endometrium was shown in randomized clinical trials of up to 52 weeks [31]. Despite the lack of adverse safety data, vaginal estrogen carried the same box warning as systemic hormone therapy. However, in November 2025, the US Food and Drug Administration (FDA) requested that manufacturers remove the boxed warning from all estrogen products, including low-dose vaginal estrogen products [32]. The change in labeling highlights that low-dose vaginal estrogen is a safe treatment option for most patients with GSM, as well as other patients with hypoestrogenism (eg, premenopausal patients with lactation-induced vaginal dryness). (See 'Other premenopausal patients with hypoestrogenism' below.)

Low-dose vaginal hormone therapies may also be an option for some patients with estrogen-sensitive cancers (eg, endometrial, breast) who fail nonpharmacologic and nonhormone treatments [33], but requires a discussion of risks and benefits with the patient and involvement of the patient's oncologist. (See 'Patients with breast cancer' below and 'Patients with other cancers' below and "Overview of approach to endometrial cancer survivors", section on 'Sexual dysfunction'.)

**Absorption by sexual partners** — Clinical implications of the absorption of vaginal estrogen by a sexual partner are generally unknown, and there are no data to support refraining from coital activity for a specific period of time after use of a vaginal estrogen [7]. Exposure of a male partner to vaginal estrogen cream during sexual intercourse is rarely clinically significant [34-36]. However, if this remains a concern for the patient, the patient can be counseled to modify the application (eg, use the vaginal estrogen cream at an earlier time in the day), so less cream is present on the vulvovaginal area during sexual activity.

**Preparations: Cream, tablet, capsule, ring** — The preparations of vaginal estrogen that are commercially available in the United States are [conjugated estrogens](#) (cream) and [estradiol](#) (cream, tablet, capsule, and ring) ( [table 3](#)). In

| Vaginal hormone preparations available in the United States for treatment of genitourinary symptoms of menopause (vulvovaginal atrophy) |                            |                                                 |                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation                                                                                                                             | United States trade name   | Available strength(s)                           | Suggested regimen                                                                                                                                                                                                      |
| <b>Estrogens</b>                                                                                                                        |                            |                                                 |                                                                                                                                                                                                                        |
| Conjugated estrogen                                                                                                                     | Vaginal cream <sup>a</sup> | 0.025 mg conjugated estrogens per g of cream    | 0.5 g of cream intravaginally administered once daily for 2 weeks, then reduce to twice weekly <sup>b</sup> . Dose may be adjusted according to patient's symptoms and clinical response (range: 0.5 to 2 g of cream). |
| Estrogen                                                                                                                                | Vaginal cream <sup>a</sup> | 0.01 mg estradiol per g of cream                | 0.5 g of cream intravaginally administered once daily for 2 weeks, then reduce to twice weekly <sup>b</sup> . Dose may be adjusted according to patient's symptoms and clinical response (range: 0.5 to 2 g of cream). |
|                                                                                                                                         | Vaginal tablet             | 0.01 mg or 0.02 mg estradiol per vaginal tablet | Insert 1 tablet intravaginally once daily for 2 weeks, followed by twice weekly, unless dose depends on clinical severity and patient preference. Dose may be increased or                                             |

Europe and some other countries, estriol cream or suppositories also are available.

A systematic review of 19 randomized trials including over 4000 patients investigated local estrogen treatment and found that creams, inserts, and rings were all similarly effective in relieving symptoms of vaginal atrophy [37].

The vaginal estrogen preparations with the lowest systemic absorption are the standard regimens of the 4 mcg or 10 mcg [estradiol](#) tablet or capsule and the 7.5 mcg/day estradiol ring [38-41].

We define low-dose vaginal estrogen as  $\leq 50$  mcg [estradiol](#) or  $\leq 0.3$  mg [conjugated estrogens](#) (in  $\leq 0.5$  g cream). Other doses of conjugated estrogens cream ( $\geq 0.625$  mg in 1 g of cream) and estradiol vaginal cream ( $>100$  mcg in  $>1$  g of cream) are considered higher dose preparations. Higher serum estrogen concentrations potentially result in endometrial proliferation [41,42]. (See 'Use of an opposing progestogen' below.)

Systemic absorption, patient preference, convenience, and cost should guide the choice of preparation [43]. In our practice, we use cream for patients who have symptomatic vulvar atrophy (eg, fissures) so that cream may also be applied to areas of the vulva affected by atrophy. When the vulvar atrophy improves, we switch to a vaginal insert or ring, depending on patient preference. For other patients, we start the vaginal insert and switch to the ring if long-term therapy is desired and if the patient is comfortable with use of the ring and cost is not a barrier.

- **Tablet or capsule** – There are two strengths of vaginal [estradiol](#) inserts in the United States: 10 mcg (Vagifem, Yuvaferm, Imvexxy) or 4 mcg (Imvexxy) [23,44,45]. A 25 mcg tablet was discontinued in 2010. The insert is placed in the vagina daily for the first two weeks of use and then twice weekly thereafter [46,47]. In our experience, thrice-weekly use may be used for patients in whom twice-weekly therapy does not provide sufficient symptom relief. For Imvexxy, either the 4 mcg or 10 mcg insert may be prescribed, and the dose adjusted up or down based on clinical response.

Randomized trials have demonstrated that the 10 mcg [estradiol](#) insert is effective for relief of vaginal symptoms [23,44,45]. Although the 10 mcg estrogen dose is significantly more effective than placebo therapy, one randomized trial found it to be slightly less effective than a 25 mcg dose for initially inducing maturation, providing relief of vaginal symptoms, decreasing pH, and improving urogenital atrophy [44]. Another randomized trial (n = 764) comparing 4, 10, or 25 mcg doses with placebo found that all three doses were more effective than placebo at treating GSM symptoms [48]. Efficacy was fairly

|                       |                |                                                   |                                                                                                                           |
|-----------------------|----------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                       |                |                                                   | Decreased according to clinical response.                                                                                 |
|                       | vaginal insert | 10 mcg estradiol per vaginal insert               | Insert 1 tablet intravaginally once daily for 2 weeks, followed by twice weekly.                                          |
|                       | vaginal ring   | 7.5 mcg estradiol per ring, released over 90 days | Ring is inserted into the vagina by the patient or clinician. Ring is removed and replaced with a new ring every 90 days. |
| estrogen <sup>a</sup> |                |                                                   |                                                                                                                           |
| Progestin             | vaginal insert | 0.01 mg medroxyprogesterone per vaginal insert    | Insert 1 suppository intravaginally once daily.                                                                           |

Vaginal hormone preparations available in the United States for treatment of genitourinary syndrome of menopause (vulvovaginal atrophy)

Table 3 - larger image below

similar across doses. Dyspareunia was improved at two weeks with all three doses. Vaginal dryness was improved at two weeks with the 10 and 25 mcg doses and at six weeks with the 4 mcg dose. Vaginal irritation and/or itching was improved at 12 weeks with the 10 and 25 mcg doses but not the 4 mcg dose.

In terms of serum estrogen absorption, in one 12-week study of postmenopausal patients with baseline serum [estradiol](#) concentrations of approximately 7 to 8 pg/mL, mean serum estradiol concentrations over a 24-hour period for 10 and 25 mcg estradiol inserts were 11 and 23 pg/mL (40.4 and 84.4 pmol/L), respectively [40]. Another randomized trial (n = 72) evaluated serum estradiol levels in postmenopausal patients using a 4, 10, or 25 mcg dose (daily for two weeks followed by twice weekly for 10 weeks) versus placebo [49]. Compared with placebo, 4 mcg showed no significant difference in any serum estradiol measures; 10 mcg showed an increase in the maximum estradiol concentration on day 1 (10.9 versus 6.6 pg/mL in patients in the placebo group) but not on day 14; and 25 mcg showed both higher maximum and average levels on both day 1 and day 14.

- **Ring** – A silastic ring impregnated with [estradiol](#) (Estring) delivers estrogen locally to the vagina [50-52]. The ring is designed to release 7.5 mcg of estradiol to the vagina daily for a period of 90 days, at which time the patient or clinician replaces it with a new ring [53].

In prospective studies of the low-dose [estradiol](#) ring, symptomatic improvement was comparable to that of orally administered estrogen, and vaginal cytology resembled that of premenopausal patients [51,52,54-58]. The ring was well tolerated and did not interfere with intercourse; however, patients who had previously undergone a hysterectomy sometimes had problems with ring retention.

The serum [estradiol](#) levels with use of the low-dose ring (7.5 mcg/day) are 5 to 10 pg/mL serum estradiol [26,38,39,59].

Of note, another vaginal ring (Femring), releases much higher doses of [estradiol](#) (50 to 100 mcg per day) and is considered systemic rather than local vaginal estrogen therapy. It is used for the treatment of both vasomotor and GSM symptoms. (See '[Patients on systemic hormone therapy for other indications](#)' below.)

- **Cream** – The effectiveness of vaginal estrogen cream for treating vaginal atrophy is well established. Studies have consistently found that in postmenopausal patients initially given 0.5 g of estrogen cream daily, vaginal cytology changed to that of normal premenopausal patients [42,50,60].

Two creams for vaginal estrogen therapy are available in the United States: [conjugated equine estrogen](#) cream and [estradiol](#) cream. The estrogen dose varies with the amount of cream used; the creams are supplied in a tube with an applicator marked with various

doses so that the patient can measure the prescribed amount [61,62]. For patients who are not sexually active, cannot, or choose not to use the cream applicators, an alternative method of delivery is the 'fingertip' method; 1 g of cream (eg, a dime-sized amount or amount equal to the length of the index fingertip) is applied to the introitus.

In our practice, we start vaginal estrogen creams with daily dosing for two weeks and then change to twice weekly. Patients often achieve relief of symptoms with doses lower than the manufacturer-recommended doses. For patients with more severe symptoms, the clinician may start with a dose on the higher end of recommended doses and then decrease the dose when symptoms improve or switch to the lower dose options, the tablet, capsule, or ring. Since the goal is symptomatic relief, after this goal is achieved, the patient is often able to determine the lowest effective dose.

The two cream preparations available in the United States contain different types of estrogens; other preparations (eg, [estrone](#) [Estragyn]) are available in Canada and Europe [63]:

- **Conjugated estrogens** (0.625 mg [conjugated equine estrogens](#)/1 g cream [42]; Premarin) – Doses range from 0.5 to 2 g of cream. This equals 0.3 to 1.25 mg of conjugated estrogens.

The regimens recommended by the manufacturer for vulvovaginal atrophy (VVA) are: 0.5 to 2 g of cream intravaginally administered in either a continuous regimen (twice weekly) or cyclic regimen (daily for 21 days then off for 7 days) [64]. We do not use a cyclic regimen in our practice.

Serum absorption is difficult to measure since [conjugated estrogens](#) contain more than 200 compounds, some of which are estrogenic, with 50 to 58 percent consisting of sodium oestrone sulfate and 25 to 35 percent consisting of sodium equilin sulfate. The plasma estradiol after use of conjugated estrogens bears no relationship to actual activity since activation of some components can take place in the tissue itself [65].

- **Estradiol** (100 mcg [estradiol](#)/1 g cream; Estrace) – Doses range from 0.5 to 4 g of cream. This equals 50 to 400 mcg of estradiol. 200 mcg of estradiol cream results in a serum estradiol level of approximately 40 pg/mL [66]. We use 0.5 g of estradiol cream, lower than recommended by the manufacturer.

The regimen recommended by the manufacturer is 2 to 4 g of cream intravaginally administered daily for one or two weeks, then gradually reduced to half the initial dose for a similar period [67]. A maintenance dose of 1 g of cream, one to three times per week, may be used.

It should be noted that the maximal doses of various forms of vaginal creams indicated on the package insert produce premenopausal plasma levels of estradiol (premenopausal [estradiol](#) ranges from 40 to 600 pg/mL, depending upon the phase of the menstrual cycle) [68] and should not be used long term without an opposing progestin and/or endometrial assessment. (See '[Use of an opposing progestogen](#)' below.)

For patients who develop sensitivity to the preservatives in the creams, preservative-free [estradiol](#) cream can be compounded, but these preparations are not FDA approved nor monitored.

- **Estriol** – Estriol vaginal cream and suppositories are available in many countries outside the United States [69]. Estriol is the 16-hydroxylated metabolite of [estradiol](#). In vulvar and vaginal tissues, estriol is a weak estrogen agonist. Estriol is not approved by the FDA for the treatment of GSM.

**Use of an opposing progestogen** — For patients with GSM treated with low doses of vaginal estrogen ( [table 3](#)), a progestogen (progestin or [progesterone](#)) is generally not indicated to protect against endometrial hyperplasia or carcinoma [68,70]. However, clinical trial data supporting endometrial safety beyond one year are lacking [31,71].

Estrogen creams that are used for systemic therapy and not local therapy for GSM may result in higher serum estrogen levels, but in our practice, we do not use an opposing progestin with [estradiol](#) cream and [conjugated estrogen](#) cream doses as prescribed for GSM. The total impact on endometrial neoplasia is likely to be impacted by dose, frequency, and duration of therapy.

As noted above, the systemic estrogen absorption of the vaginal creams is difficult to quantify precisely due to limitations of assays and the difficulty in measuring the total serum estrogen level of the multiple estrogens in [conjugated estrogen](#) cream. There are few data regarding the risk of endometrial neoplasia with these preparations, and there are no studies of the impact of low-dose estrogen on endometrial activity after more than one year of treatment [31,72].

Studies that have evaluated the endometrial effect of the vaginal estrogen preparations by ultrasonography or by biopsy include the following:

- For the 10 mcg [estradiol](#) insert (Vagifem), a prospective study evaluated daily use for two weeks then twice weekly for 52 weeks [73]. Among the 386 patients who had an evaluable biopsy 52 weeks after beginning therapy, there was one case of complex hyperplasia without atypia in a patient exposed to the drug for only nine days (suggesting a preexisting abnormality) and one case of endometrioid adenocarcinoma (in whom preexisting disease could not be excluded because the pretreatment biopsy sample was

inadequate for analysis). A previous study had supported the endometrial safety of the 25 mcg estradiol vaginal insert, which has been discontinued [39].

- For the low-dose [estradiol](#) ring (Estring), 60 postmenopausal patients were randomly assigned to receive Estring or no treatment for 12 months [38]. Sonographic evaluation found no significant increases in endometrial thickness in either group.
- For low-dose [conjugated estrogens](#) cream (Premarin), a study of 20 participants (treated with 0.3 mg of conjugated estrogens, administered vaginally three nights per week for six months) found evidence of endometrial hyperplasia on biopsy in one, even though the sonographic endometrial thickness was normal [74]. This observation suggests caution, but further study is needed.

Regardless of the estrogen dose, patients with postmenopausal bleeding, including those who report this while using vaginal estrogen, should be evaluated, as appropriate. (See ["Approach to the patient with postmenopausal uterine bleeding"](#).)

**Response and duration of therapy** — In our experience, patients usually have improvement in symptoms after two to four weeks of vaginal estrogen therapy. In a trial comparing the 4 and 10 mcg dose of [estradiol](#) vaginal tablets, a positive response by two weeks of treatment was indicative of continued response at 12 weeks [75].

The duration of treatment needed to improve symptoms and maintain this improvement may vary across patients and should be individualized according to each patient's degree of vaginal atrophy symptoms. Low-dose vaginal estrogen therapy may be used indefinitely, based upon the low risk of adverse effects, although clinical trials to date have not followed patients beyond one year. The duration of higher dose vaginal therapy or systemic estrogen therapy should be guided by the risks and benefits of therapy.

**Patients on systemic hormone therapy for other indications** — For patients receiving systemic estrogen therapy for other menopausal effects (eg, vasomotor symptoms), low-dose vaginal therapy may be **added** if relief of atrophic symptoms is insufficient. Prevention/treatment of osteoporosis is another benefit of systemic hormone therapy, although estrogen is not considered first-line therapy for this. Low-dose vaginal estrogen is unlikely to be effective in osteoporosis, although one study reported a reduction in bone resorption in postmenopausal patients receiving low-dose vaginal [estradiol](#) (7.5 mcg/24 hours), which suggests that a systemic effect is possible [76].

**Other medications** — There are alternative hormonal medications other than estrogen that are used to treat GSM. [Vaginal DHEA](#) may result in decreased exposure to estrogen, but there appears to be some increase in serum estrogen levels even with this formulation. Some data suggest an improvement in libido with vaginal DHEA, but there are few data and the effect, if present, is modest. Oral [ospemifene](#), a selective estrogen receptor modulator (SERM), is

another option for patients who prefer oral medications or who are not able to insert a vaginal medication. Some clinicians prescribe vaginal [testosterone](#) off-label for GSM. This is usually with the concurrent goal of treating low libido.

**Dehydroepiandrosterone (prasterone)** — [Vaginal DHEA](#), also referred to as prasterone, is a treatment option for dyspareunia associated with GSM. Prasterone is effective but is associated with a slight increase in circulating DHEA, [testosterone](#), and [estrone](#) levels, and its efficacy has not been compared directly with vaginal estrogen [18,77,78]. In 2016, the FDA approved the use of prasterone 6.5 mg (0.5% formulation) as a daily vaginal suppository for the treatment of dyspareunia in the setting of vulvar or vaginal atrophy due to menopause [79]. In systematic reviews of randomized trials and cohort studies evaluating peri- and postmenopausal patients with GSM, vaginal DHEA resulted in improvements in vaginal maturation index (VMI), vaginal dryness, dyspareunia, sexual function, and lowered vaginal pH [10,18].

For most patients with symptomatic VVA who do not improve with vaginal lubricants and moisturizers and who are candidates for hormonal therapy, we suggest low-dose vaginal estrogen therapy rather than [vaginal DHEA](#). There are more data and clinical experience regarding vaginal estrogen, and the twice weekly dosing is likely to be easier for most patients than the daily dosing required with DHEA. For patients with no strong contraindications to estrogen therapy, but who value avoiding estrogen more than dosing convenience, vaginal DHEA is a reasonable option.

The mechanism of action of prasterone on the vagina is likely due to the aromatization of androstenedione and [testosterone](#) locally to [estrone](#) and [estradiol](#) through a process referred to by some investigators as intracrinology [80]. In one study, weeks of [vaginal DHEA](#) (6.5 mg) caused estrone to rise from 15 to 18 pg/mL and estradiol from 2 to 3 pg/mL [77]. It has also been proposed that testosterone acting through the androgen receptor is a possible mechanism of action, but this is only based on rodent data and may not apply to people.

Prasterone was found to be an effective treatment for dyspareunia in a randomized trial that assigned 482 postmenopausal patients with dyspareunia to prasterone or placebo for 12 weeks and found improved Female Sexual Function Index (FSFI) overall scores and in the pain and satisfaction domains [81]. At 12 weeks, treatment with [vaginal DHEA](#) resulted in a significant increase from baseline in serum DHEA, [testosterone](#), and [estrone](#) but not [estradiol](#) levels, although DHEA, estradiol, estrone, and testosterone remained within the range of an average postmenopausal patient [82].

The small elevations of [estrone](#) in patients treated with prasterone raise concern for use of this treatment in patients with or at risk for estrogen-sensitive malignancies, particularly for

those with breast cancer who are treated with an aromatase inhibitor (AI). (See '[Patients with breast cancer](#)' below.)

DHEA has been studied as a potential medication to increase libido in postmenopausal patients, but its efficacy is uncertain, and the serum DHEA levels with this dose of [vaginal prasterone](#) are unlikely to have an impact. Also, libido often improves with any adequate treatment of dyspareunia and is likely due to a reduction in pain or discomfort and increased blood flow and lubrication.

**Ospemifene** — [Ospemifene](#) is a SERM that acts as an estrogen agonist in the vagina and appears to have no clinically significant estrogenic effect on the endometrium or breast. Ospemifene was approved by the FDA in 2013 for the treatment of moderate to severe dyspareunia and/or vaginal dryness caused by VVA in menopausal patients [83].

For patients with symptomatic VVA that is not relieved with nonpharmacologic therapy, we suggest [ospemifene](#) rather than vaginal estrogen therapy for those who cannot (severe arthritis, obesity, vulvodynia) or prefer not to use a vaginal product. The disadvantages of ospemifene compared with vaginal estrogen are need for daily use and systemic side effects (hot flashes, potential risk of thromboembolism). The safety of ospemifene has not been demonstrated in patients with a prior history or an increased risk of breast cancer or in patients with an increased risk of thromboembolism.

[Ospemifene](#) is effective in treating dyspareunia and vaginal dryness in menopausal patients with VVA compared with placebo; no studies to date have directly compared ospemifene with vaginal estrogen therapy. Two randomized trials found that ospemifene (60 mg/day, oral, 12 weeks of therapy) was significantly more effective than placebo in improving dyspareunia, although the benefit beyond that of placebo was modest [84,85]. Severity of dyspareunia was rated with a four-point scale; in one trial, dyspareunia decreased by two to three points in 53 percent of participants in the treatment group compared with 39 percent of those who received placebo [85]. The proportion of those for whom dyspareunia was absent or mild was 38 and 25 percent, respectively, compared with 28 and 19 percent in the placebo group. In addition, either 30 or 60 mg of ospemifene was found to significantly decrease vaginal dryness in one trial [84]. Another randomized trial of 919 patients found that ospemifene (60 mg/day, 12 weeks) resulted in improvements in the sexual pain, arousal, and desire domains of the FSFI, a validated measure of female sexual function [86]. Similarly, a trial comparing 12 weeks of oral ospemifene 60 mg with placebo reported improved vulvar-vestibular health indices and FSFI scores in the ospemifene group [87]. In the systematic reviews of peri- and postmenopausal patients described above (see '[Dehydroepiandrosterone \(prasterone\)](#)' above), ospemifene lowered vaginal pH; decreased urinary urgency, nocturia, and urinary tract infections; and improved dryness, dyspareunia, and sexual function using FSFI [10,18]. However, rates of vasomotor symptoms, vaginal

candidiasis, and endometrial thickening were increased; no patients developed endometrial hyperplasia at one year.

Hot flushes are the most common adverse effect of [ospemifene](#) [85,88]. In one trial, the rate of hot flushes was 6.6 percent in the treatment group compared with 4.3 in the placebo group; however, symptoms typically resolved by six months and led to discontinuation in only two patients [85]. Ospemifene appears to have a favorable endometrial safety profile. Ospemifene results in an increase in endometrial thickness (0.4 to 0.7 mm in 12 weeks [84,85,89]), but studies of one year of use have found no cases of atypical endometrial hyperplasia or endometrial carcinoma [90,91].

Thrombotic adverse effects are a potential risk with SERMs. There have been no reports of such complications with [ospemifene](#), but more data are needed to detect this relatively rare complication.

Animal and preclinical data suggest that [ospemifene](#) has a neutral or inhibitory effect on carcinogenesis in the breast [92,93]. As an example, one tissue culture study found that ospemifene had an antiproliferative effect on human breast tissue but was less potent an inhibitor than [tamoxifen](#) or [raloxifene](#) [94]. Further study is needed to evaluate the safety of ospemifene in patients with breast cancer or whether it may have a protective effect for patients at high risk for breast malignancy.

A potential benefit of [ospemifene](#) is a reduction in bone turnover [95,96]. Use of SERMs for the prevention and treatment of osteoporosis is discussed in detail separately. (See "[Selective estrogen receptor modulators for prevention and treatment of postmenopausal osteoporosis](#)".)

**Testosterone** — Existing data provide some support for the concept that vaginal [testosterone](#) may stimulate vaginal mucosa and improve symptoms of vaginal atrophy [97]. However, none are FDA-approved for use in females [98]. Some providers may prescribe vaginal testosterone to patients with GSM who also desire treatment with testosterone for low libido. There has also been interest in using vaginal testosterone for GSM in patients on AI therapy, but the safety of this approach remains uncertain. (See "[Overview of sexual dysfunction in females: Management](#)", section on 'Androgens' and 'Patients with breast cancer' below.)

Several studies have reported therapeutic effects of [testosterone](#) on the vaginal epithelium, including:

- Studies have reported the presence of androgen receptors in vaginal tissue and an inverse correlation with age [99,100].

- Two studies examined the effect of vaginally administered androgen on vaginal maturation parameters in patients receiving AIs. In these studies, blockade of conversion of androgen to estrogen allowed one to conclude that the effects were androgen-mediated. In one study of 21 patients, the vaginal maturation index increased, and dyspareunia and vaginal dryness improved [101]. In another study, for 33 patients in the vaginal [testosterone](#) arm, the score for rugae, petechiae, elasticity, and dryness improved, but serum testosterone increased from a mean of 33 ng/100 mL to 171 ng/100 mL at 12 weeks [102].

**Tibolone** — [Tibolone](#) is a synthetic steroid whose metabolites have estrogenic, progestogenic, and androgenic properties. It is not available in the United States, but is used in Europe and other countries to prevent hot flashes, bone loss, and vaginal atrophy and has variable effects on sexual function [103-105]. It is a systemic therapy, with a higher rate of unscheduled bleeding than placebo, but does not require concurrent progestogen therapy. (See "[Overview of sexual dysfunction in females: Management](#)", section on '[Tibolone](#)' and "[Preparations for menopausal hormone therapy](#)", section on '[Tibolone](#)'.)

[Tibolone](#) is contraindicated in patients with breast cancer; adverse effects are discussed separately. (See "[Preparations for menopausal hormone therapy](#)", section on '[Tibolone](#)'.)

## Therapies needing further evaluation

**Laser or radiofrequency devices** — Laser and other energy-based devices (eg, fractional microablative CO<sub>2</sub> laser, erbium:YAG [Er:YAG] laser, temperature-controlled radiofrequency [RF] laser) have been marketed for treatment of VVA, but there are limited data regarding the safety and efficacy of these devices. As the quality of available evidence is considered "very low" or "low" [106], we do not perform later treatment for VVA in our practice. In a properly counseled patient who has no other options, this may be considered for treatment of GSM but **not** for other indications (eg, sexual enhancement, lichen sclerosis, vulvar vestibulitis syndrome) without more data.

Laser technologies deliver fractional CO<sub>2</sub> laser energy, and some systems use nonablative photothermal Er:YAG laser to the vaginal wall tissue. Laser therapy typically consists of three laser treatment sessions over a specified time period (usually one session every four to six weeks). Beneficial effects appear to result from remodeling of vaginal tissue [107]. However, optimal patient inclusion or exclusion criteria, treatment number, and maintenance therapy are not yet known [108].

Laser or energy-based devices have not been cleared or approved by the FDA for the treatment of VVA. In July 2018, the FDA issued a safety communication warning patients about the risks associated with use of these devices, which include vaginal burns, scarring, pain during sexual intercourse, and recurring/chronic pain [109]. In August 2018, the American College of Obstetricians and Gynecologists advised that additional data from

randomized trials are needed to further assess the efficacy and safety of this procedure [110]. A 2020 clinical consensus statement by the American Urogynecologic Society also noted that, while energy-based therapies had shown treatment promise, long-term outcomes were not yet understood [108].

Randomized trials on laser treatment for GSM are limited [111-113]:

- In a multicenter randomized trial of 69 menopausal patients comparing treatment of vaginal dryness with either vaginal laser or estrogen cream, patient-reported symptoms of vaginal dryness, sexual function, and urinary distress improved, but results were similar between groups [112]. Vaginal maturation index scores, an objective measure of vaginal health, were higher after treatment in the estrogen group. Overall patient satisfaction scores were between 70 and 85 percent in both groups, and no serious adverse effects were noted. One major limitation was the early closure of this trial due to the FDA imposing an investigational device exemption, which was not obtained.
- In a subsequent randomized trial including 85 postmenopausal patients with vaginal symptoms, patients treated with CO<sub>2</sub> laser or sham therapy experienced improvement in symptom severity (as assessed on visual analog scale) at 12 months, but the magnitude of improvement was similar between groups; both participants and outcome assessors were masked to group assignment [111]. Quality of life scores and histologic changes on vaginal biopsy were also similar. Adverse events occurred in both groups and included vaginal pain, lower urinary tract symptoms, and vaginal discharge; no severe adverse events were reported. These results were supported in an animal study including 24 ewes with surgically induced menopause and vaginal atrophy [114]. In this randomized trial, more animals treated with an [estradiol](#) implant demonstrated a reversal of vaginal atrophy (ie, increase of epithelial thickness, glycogen-positive cell layers, and lubrication) compared with those treated with Er:YAG or sham laser. The response in ewes treated with YAG and sham laser were similar.
- In a randomized trial including 43 patients with a history of breast cancer and urogenital atrophy, those treated with intravaginal laser therapy (two courses per month) or hyaluronic acid suppositories (three times per week continuously for three months) had similar improvements in their vaginal health index (VHI) at three months [115].
- In a randomized trial including 49 postmenopausal patients with vaginal symptoms, treatment with CO<sub>2</sub> laser versus sham therapy resulted in similar symptom severity, as assessed by visual analog scale and Vulvovaginal Symptom Questionnaire [116]. Change in vaginal histology, which is a common surrogate determinant of treatment success, was also similar in both groups at six months postprocedure.

Systematic reviews (including randomized trials and/or observational studies) and a prospective feasibility study have reported that, overall, laser intervention appears to be a

safe and potentially effective nonpharmacologic intervention for GSM [10,106,117-119]. In one meta-analysis of 25 studies including 1152 patients with GSM, vaginal laser treatment reduced VVA and/or GSM symptoms (dryness, dyspareunia, itching, burning, dysuria); while there was a trend toward improved FSFI, this reached clinical significance in two studies [106]. However, the quality of these studies are low or very low [120]. Additional large randomized clinical trials are needed to determine the benefits, risks, and cost-effectiveness of laser therapy for VVA.

### **Alternative and complementary therapies**

- Oral vitamin D and vaginal vitamin E have been proposed, but efficacy data are limited and/or discordant [121-124].
- Supplements containing phytoestrogens (eg, soy, fenugreek) or mixed herbal and botanical preparations have been described [124].
- Oral and vaginal probiotics to alter the vaginal microbiota (eg, increase in *Lactobacillus* species) could possibly be beneficial for the treatment of symptoms of GSM, but comprehensive trials are needed for validation, and there are currently no probiotic therapies that have been validated to be an effective management intervention for GSM [124-126].

While topical vaginal estrogen is associated with a change in microbiome composition (eg, increased dominance of *Lactobacillus*) [127], and a combination of vaginal low-dose estriol and lactobacilli is an approved therapy in many countries (not in the United States), further study of this combination is needed [128].

- Cannabidiol (CBD) is an anti-inflammatory and vasodilator. While it is feasible that topical CBD could be useful in patients with GSM, there are no adequate studies to address the use of topical, vaginal, or systemic CBD for such patients. However, as some patients may be using it, inquiry about its use is appropriate. Use of CBD in the medical setting is discussed separately. (See "[Medical use of cannabis and cannabinoids in adults](#)".)

Patients who use an alternative or complementary treatment should check with the manufacturer regarding whether the product contains estrogen or other hormones.

**No role for systemic hormone therapy alone** — Systemic hormone therapy is not used for the management of vulvovaginal atrophy, unless other symptoms of menopause (eg, vasomotor symptoms) are present [2,70]. (See '[Patients on systemic hormone therapy for other indications](#)' above.)

The preference for vaginal estrogen is based on its superior efficacy for GSM symptoms and the risks of systemic menopausal hormone therapy. A meta-analysis of 58 comparative studies of patients with urogenital atrophy found that patients' reports of symptom relief

were significantly higher for vaginal versus oral estrogen therapy [129]. The reported efficacy rate is approximately 80 to 90 percent for vaginal therapy and 75 percent for systemic therapy, based upon observational data [52,130-133]. In addition, the increased risks of breast cancer and thrombosis with systemic estrogen and progestin compared with placebo were best demonstrated by the Women's Health Initiative randomized trial; some of these risks vary depending upon whether a progestin is used in combination with estrogen [134]. (See "[Menopausal hormone therapy: Benefits and risks](#)".)

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## SYMPTOMS REFRACTORY TO HORMONE THERAPY

Given the proven success of vaginal estrogen therapy, treatment failure warrants evaluation for other etiologies of the patient's symptoms. Prior to proceeding with further evaluation, compliance with therapy should be confirmed. In addition to reviewing the treatment plan with the patient, an additional option is to check a vaginal maturation index and pH to assess for adequate response to hormonal treatment. (See "[Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms](#)", section on 'Limited role of laboratory evaluation'.)

**Evaluate for other etiologies** — Vulvovaginal pain, dyspareunia, urinary incontinence, dysuria, or urinary frequency occur with GSM and may also be associated with vulvodynia, pelvic floor dysfunction, or bladder pain syndrome. Patients should be evaluated for these conditions and receive appropriate therapy. (See "[Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms](#)", section on 'Differential diagnosis'.)

A common clinical scenario is continued vulvovaginal discomfort and/or dyspareunia due to co-occurrence of GSM and vulvodynia. The vulvodynia may not have been suspected during the initial evaluation, but once the atrophy is managed, pain may continue from the vulvodynia.

These conditions are discussed in detail separately. (See "[Vulvar pain of unknown cause \(vulvodynia\): Clinical manifestations and diagnosis](#)" and "[Myofascial pelvic pain syndrome in females: Clinical manifestations and diagnosis](#)" and "[Interstitial cystitis/bladder pain syndrome: Clinical features and diagnosis](#)".)

**Pelvic physical therapy** — Pelvic physical therapy is a component of therapy in many pelvic pain syndromes and also for urinary incontinence and pelvic organ prolapse. Patients with GSM who do not respond to other therapies or who have contraindications to hormone therapy can be referred for evaluation and treatment by a pelvic physical therapist. Levator spasm often occurs after dyspareunia and can be improved with pelvic floor therapy and/or

vaginal dilators after atrophy has been treated. (See ["Female urinary incontinence: Treatment"](#), section on 'Pelvic floor muscle (Kegel) exercises'.)

**Vaginal dilators** — Vaginal dilators are a component of pelvic physical therapy. This may be particularly effective for patients who avoid intercourse due to pain since vaginal dilators are available in sets of graduated sizes and a narrow dilator can be used initially. Vaginal dilator use can often be taught to patients by their clinician, or they can be referred to a pelvic physical therapist.

In our practice, we use vaginal dilators for patients with introital stenosis or vaginal shortening/stenosis. Dilators may also be helpful for patients with GSM who have contraindications to estrogen therapy and have failed moisturizers and lubricants. Some data suggest that dilators improve vaginal function [135]. However, in patients with vaginal atrophy, dilators are likely to be more effective after hormonal treatment as elasticity within the vaginal and introitus improve. For patients with levator spasm, dilators can also be used to help relax the spasm; if unsuccessful with dilators at home, referral to pelvic floor therapy may be very helpful.

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## SPECIAL CONSIDERATIONS

**Asymptomatic patients** — Treatment of asymptomatic patients for vaginal atrophy is indicated in some clinical contexts, including:

- **Prior to vulvovaginal surgery** – Systemic or vaginal estrogen therapy is given with the goal of optimizing the identification of tissue planes and promoting wound healing. There are few data regarding this use, and it is employed depending upon surgeon preference [136,137].
- **Prior to initiating sexual activity** – For asymptomatic patients who are not sexually active but have physical examination findings of vaginal atrophy, vaginal estrogen therapy may be initiated prior to engaging in vaginal intercourse.
- **Worsening examination findings** – Asymptomatic patients with evidence of worsening vulvovaginal atrophy (VVA) on examination should be informed of these findings and preventive measures should be offered to prevent future issues (eg, recurrent vaginitis, urinary tract infection, painful intercourse). In addition, there is some evidence that early treatment with estrogen may be more effective than treatment when vulvovaginal atrophy is more severe [138].

**Patients with pelvic organ prolapse** — Some clinicians treat patients with pelvic organ prolapse, with or without symptoms, with vaginal estrogen therapy, although the benefit of this approach is uncertain. However, when pessary use is employed in the management of

these conditions, vaginal estrogen therapy is beneficial to prevent vaginal abrasions caused by the pessary. (See ["Pelvic organ prolapse in females: Epidemiology, risk factors, clinical manifestations, and management"](#), section on 'Estrogen therapy'.)

**Patients with severe anatomic changes** — Hypoestrogenism may cause moderate to severe anatomic changes in the vulva and vagina. These may include labia majora agglutination, introital narrowing, vaginal shortening or stenosis, and changes in the vaginal vault (cervix becomes flush with the vagina, fornices are obliterated).

Patients with abnormal findings on examination should be carefully examined for lesions that raise suspicion of a vulvovaginal dermatosis or malignancy, which can also cause anatomic distortion. The patient should be asked about postmenopausal bleeding and evaluated, as appropriate. (See ["Vulvar lesions: Differential diagnosis of vesicles, bullae, erosions, and ulcers"](#) and ["Approach to the patient with postmenopausal uterine bleeding"](#).)

For patients with severe atrophic changes on examination, the first aspect of care is to ask the patient about symptoms and goals. If the patient is asymptomatic and reports no need for intervention, treatment is not necessary unless urinary flow is impeded, or the patient develops vaginitis or frequent urinary tract infections; these risks should be discussed with the patient.

**Labial agglutination** — For patients with labial agglutination who are symptomatic or who want to resume sexual activity with vaginal penetration, first-line therapy is to treat with vaginal estrogen cream for four to six weeks (using a typical regimen of daily for two weeks followed by twice weekly). The patient should be instructed to apply the vaginal estrogen to the area of agglutination and also to apply gentle digital pressure. However, in cases in which a plane cannot be easily created by gentle digital pressure, surgical management with lysis of adhesions is an option and should be performed by a gynecologic surgeon experienced with this procedure. Following lysis of adhesions, use of vaginal dilators and topical estrogen, with or without a corticosteroid cream, may decrease the risk of adhesion reformation.

**Vaginal narrowing or shortening** — For patients with symptomatic introital narrowing and/or vaginal shortening or stenosis, first-line therapy is vaginal estrogen therapy and use of graduated vaginal dilators; an expanding dilator may also be used (eg, Milli). Estrogen cream may be applied to the dilator. If there is no improvement in four to six weeks, surgical options should be discussed with the patient, and surgery should be performed only by a gynecologic surgeon experienced with surgical treatment of this issue [139]. (See ["Vaginal dilators"](#) above.)

**Patients with breast cancer** — Symptoms of GSM are highly prevalent in breast cancer survivors because of the use of antiestrogen cancer therapies and the onset of treatment-related menopause. As an example, a study of 97 patients with breast cancer reported

moderate or severe symptoms of vaginal atrophy in 58 percent of patients on aromatase inhibitors and 32 percent of those on [tamoxifen](#) [140].

- Irrespective of breast cancer treatment, for patients with breast cancer with GSM symptoms, nonhormonal options (eg, lubricants, moisturizers) are **first-line therapy** [16,141,142] (see '[Initial therapy with moisturizers and lubricants](#)' above). In one single-arm prospective study including 101 postmenopausal patients with hormone receptor-positive cancer, treatment with a vaginal moisturizer (Hyalo GYN) for 12 weeks improved vulvar and vaginal symptoms in 88 and 92 percent of patients, respectively; more frequent dosing (three to five versus one to two times per week) may be required compared with postmenopausal patients without cancer [142].

When subsequent-line therapy is needed, decisions regarding treatment are guided by the type of breast cancer treatment the patient is receiving. In general, there are insufficient data regarding the safety of hormonal therapy for the treatment of GSM in females with breast cancer [7].

- For patients with breast cancer treated with selective estrogen receptor modulators, patients who have completed a full course of adjuvant endocrine therapy, or for those with hormone receptor-negative disease:
  - Low-dose vaginal estrogen or vaginal dehydroepiandrosterone (DHEA; prasterone) are likely reasonable subsequent-line choices if symptoms do not improve with nonhormonal treatments. In a meta-analysis of eight observational studies including breast cancer survivors, vaginal estrogen was not associated with an increased risk of breast cancer recurrence (six studies; 24,060 patients), breast cancer-specific mortality (four studies; 61,695 patients), or overall mortality (five studies; 59,724 patients) [30].

Vaginal estrogen and DHEA are discussed in detail above. (See '[Vaginal estrogen therapy](#)' above and '[Dehydroepiandrosterone \(prasterone\)](#)' above.)

- However, these medications should be initiated only after consultation with the patient's oncologist and after comprehensive counseling of the patient about potential risks. Limited data suggest that breast cancer survivors derive the same type of benefits as other patients when treated with intravaginal estrogen products. Furthermore, use of low-dose vaginal estrogen therapy has not been found to be associated with an increased risk of breast cancer recurrence [106,143,144]. However, studies have shown that there can be low levels of systemic absorption of estrogens in breast cancer survivors using intravaginal estrogens, which may decline as the GSM resolves [143,144].

Effects of systemic hormone replacement in breast cancer survivors is discussed elsewhere. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Breast cancer patients](#)'.)

- For patients with breast cancer treated with aromatase inhibitors (AIs):
- For most patients, we suggest **not** using vaginal estrogen, DHEA (prasterone), or off-label use of [testosterone](#). However, for patients with marked symptoms of GSM, and who are willing to accept that vaginal hormonal therapy may limit the effectiveness of their cancer treatment, short-term exposures are unlikely to affect long-term cancer outcomes. For such patients, low-dose vaginal estrogen or prasterone is reasonable. Alternatively, some oncologists may switch treatment from AI therapy to [tamoxifen](#), depending upon the risk features of the cancer and patient preferences, because of the lower risk of developing GSM.

The rationale for caution with vaginal hormones is that the aim of AI therapy is to maximally reduce serum estrogen levels, which may be slightly increased with topical hormones. However, we recognize that data are limited and that survival outcomes and recurrence rates among patients on AIs and topical hormone preparations are not available [145].

- There are limited data regarding the use of vaginal estrogen therapy in patients with early hormone receptor-positive breast cancer receiving AI therapy [102,146]. In one randomized trial of 76 patients with early-stage breast cancer taking AIs, vaginal estrogen (in the form of the [estradiol](#) ring) resulted in transient elevations of serum estradiol levels (>10 pg/mL) in 11 percent of patients [102]. No patients had persistent elevation of serum estradiol levels. Vaginal atrophy and sexual interest and dysfunction improved in all patients.

In the same Danish retrospective study discussed above, of the 403 patients with breast cancer treated with AI therapy, those receiving vaginal estrogen (23 percent) compared with no vaginal estrogen had higher rates of breast cancer recurrence (adjusted hazard ratio [HR] 1.39, 95% CI 1.04-1.85) during the 9.8 years (median) of follow-up [147]. Vaginal estrogen was not associated with an increase in mortality, which was similar between groups. There are several limitations to this study, including that only a minority (4.8 percent) of patients were treated with AI therapy and these patients may have been at higher risk for recurrence compared with those chosen for [tamoxifen](#) (or no) therapy [148]. In addition, no data were available regarding the type or dose of vaginal estrogen used, which has decreased significantly over time (see '[Preparations: Cream, tablet, capsule, ring](#)' above). Similarly, in a subgroup analysis of a claims-based study including patients with breast cancer and GSM, vaginal estrogen therapy plus concurrent AI use (90 patients) compared with no AI use was associated with higher rates of breast cancer recurrence; increased rates were observed in patients with any ER status (78 versus 16 percent, relative risk [RR] 5, 95% CI 3.1-8.2) and those with ER+ status (76 versus 29 percent, RR 2.6, 95% CI 1.6-4.5) [149]. Time to recurrence in the concurrent AI group was 140 days (average). One limitation of this study is that it did

not control for high-risk features (eg, breast cancer stage and grade). However, data are conflicting, and in the cohort study of females aged 40 to 79 years with newly diagnosed breast cancer discussed above, cancer-specific mortality (when adjusted for cancer stage and grade) was not higher in those treated with an AI plus vaginal estrogen compared with those treated with an AI alone [150].

- Off-label vaginal [testosterone](#) therapy has also been evaluated based on the hypothesis that testosterone therapy should not increase serum estrogen levels [101,102,151,152]. However, in the randomized trial detailed above, vaginal testosterone resulted in both transient and persistent elevations of serum [estradiol](#) levels (12 percent of patients in both groups); vaginal atrophy and sexual interest and dysfunction improved in all patients [97,102].
- [Vaginal DHEA](#) therapy (prasterone) in patients with breast (97 percent) or gynecologic cancer was evaluated in a randomized trial of 464 postmenopausal patients, approximately 60 percent of whom were on AI therapy [153]. Participants received 12 weeks of treatment with either 3.25 mg DHEA, 6.5 mg DHEA, or nonhormonal vaginal moisturizer. Patient-rated severity of dryness or dyspareunia was similar across all three groups, although those on the higher dose of DHEA reported better sexual function as measured by the Female Sexual Function Index. Circulating DHEA-sulfate and [testosterone](#) levels were increased in the vaginal DHEA groups [154]. [Estradiol](#) was slightly increased with DHEA 6.5 mg/day (from an average of 3.6 pg/mL to 4 pg/mL) but not with 3.25 mg/day.

Other therapies have been proposed to treat GSM in breast cancer survivors. Use of estriol rather than [estradiol](#) has been suggested for patients with breast cancer since its metabolic clearance is more rapid [145]. Vaginal estriol preparations are available in many countries but not in the United States. However, while clearance is more rapid, if used in a manner in which serum levels are consistently elevated, estriol can act as a systemic estrogen; therefore, the same cautions as vaginal estradiol use are applied here.

[Pilocarpine](#), a cholinergic agonist used to treat Sjögren's disease, was investigated for this indication but was **not** found to improve vaginal dryness compared with placebo in a randomized trial of 201 patients [155].

Fractional CO<sub>2</sub> laser therapy has also been studied in breast cancer survivors and may improve GSM symptoms [115,119]; however, the safety and efficacy of these devices remain uncertain, as discussed in detail above. (See '[Laser or radiofrequency devices](#)' above.)

**Patients with hereditary cancer syndromes** — The management of GSM in patients with hereditary cancer syndromes (eg, breast cancer type 1 or 2 susceptibility genes [*BRCA1* and *BRCA2*]) without a cancer diagnosis is discussed in detail separately. Genetic consultation is also recommended. (See "[Cancer risks and management in BRCA1/2 carriers](#)", section on

'Vaginal estrogen therapy' and "Genetic testing for hereditary breast and ovarian cancer syndromes".)

**Patients with other cancers** — Symptoms of GSM are also highly prevalent in survivors of other cancers. In addition to nonpharmacologic and nonhormone treatments, low-dose vaginal hormone therapies are often an option for such patients. Counseling requires a discussion of risks and benefits with the patient and may include involvement of the patient's oncologist. This is discussed in detail separately. (See "[Overview of sexual dysfunction in female cancer survivors](#)", [section on 'Management'](#)".)

**Other premenopausal patients with hypoestrogenism** — Other premenopausal patients with hypoestrogenism, including those who are postpartum and/or lactating, may also experience GSM symptoms. In addition to nonpharmacologic and nonhormone treatments, low-dose vaginal hormone therapies are often an option for such patients. (See '[Initial therapy with moisturizers and lubricants](#)' above and '[Persistent symptoms](#)' above and "[Overview of the postpartum period: Disorders and complications](#)", [section on 'Vaginal dryness'](#)".)

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## SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Menopause](#)".)

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## INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5<sup>th</sup> to 6<sup>th</sup> grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10<sup>th</sup> to 12<sup>th</sup> grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Vaginal dryness \(The Basics\)](#)")
- Beyond the Basics topics (see "[Patient education: Vaginal dryness \(Beyond the Basics\)](#)")

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## SUMMARY AND RECOMMENDATIONS

- **Clinical approach** – Genitourinary syndrome of menopause (GSM) is defined as a collection of symptoms and signs (eg, dryness, burning, dyspareunia) related to a decrease in estrogen and other sex steroids. GSM affects the labia majora/minora, clitoris, vestibule/introitus, vagina, urethra, and bladder. Clinicians should ask hypoestrogenic patients about these symptoms as many do not discuss them on their own. Management is guided by the patient's symptoms and goals. (See '[Clinical approach](#)' above.)
- **Initial therapy** – For most patients with vulvovaginal symptoms due to GSM, nonhormonal vaginal moisturizers and lubricants are the preferred approach to initial therapy ( [table 1](#) and [table 2](#)). Pelvic floor muscle (Kegel) exercises may also be encouraged; a referral to a pelvic floor therapist can often be helpful. (See '[Initial therapy with moisturizers and lubricants](#)' above.)
- **Subsequent therapy for most patients** – For most patients with vulvovaginal symptoms due to GSM who are not adequately treated with vaginal moisturizers and lubricants, we suggest low-dose vaginal estrogen rather than dehydroepiandrosterone (DHEA; prasterone) or [ospemifene \(Grade 2C\)](#). (See '[Patients on systemic hormone therapy for other indications](#)' above.)
  - Available vaginal estrogen preparations include [conjugated estrogens](#) (cream) and [estradiol](#) (cream, tablet, capsule, and ring) ( [table 3](#)). In Europe and some other countries, estriol cream or suppositories are also available. All preparations are effective in relieving symptoms of vaginal atrophy. Systemic absorption, patient preference, convenience, and cost should guide the choice of preparation. (See '[Preparations: Cream, tablet, capsule, ring](#)' above.)
  - In our practice, we use cream for patients who have symptomatic external atrophy (eg, vulvar fissures) so that the cream may also be applied directly to these areas. When the vulvar atrophy improves, we switch to a vaginal insert or ring, depending on patient preference. For patients without external atrophy, we start with a vaginal insert and switch to the ring if long-term therapy is desired and if the patient is comfortable with use of the ring. (See '[Preparations: Cream, tablet, capsule, ring](#)' above.)
- **Alternative therapies for most patients**
  - For most patients with GSM who want to avoid using vaginal estrogen but have no contraindications to estrogen itself and no objections to vaginal therapy, [vaginal DHEA](#) (prasterone) is an option. One disadvantage is the frequent dosing schedule (daily as compared with twice weekly for vaginal estrogen). While DHEA is approved only for the indication of GSM-associated dyspareunia, studies have also demonstrated improvement

in vaginal dryness. Vaginal DHEA has not been thoroughly studied in patients with a history of breast or gynecologic cancer. (See '[Dehydroepiandrosterone \(prasterone\)](#)' above.)

- For most patients with GSM who cannot or prefer not to use a vaginal product, oral [ospemifene](#) is an option. Disadvantages of ospemifene include daily use and systemic side effects (eg, hot flashes, potential risk of thromboembolism). (See '[Ospemifene](#)' above.)
- Some clinicians prescribe vaginal [testosterone](#) off-label for GSM. This is usually with the concurrent goal of treating low libido. (See '[Testosterone](#)' above.)
- Systemic hormone therapy is **not** used for the management of vulvovaginal atrophy, unless other symptoms of menopause (eg, vasomotor symptoms) are present. (See '[No role for systemic hormone therapy alone](#)' above.)
- **Subsequent therapy for patients with breast cancer** – For patients with breast cancer in whom symptoms of GSM are not adequately managed with nonhormonal vaginal moisturizers and lubricants, management is guided by the type of breast cancer treatment the patient is receiving:
  - For patients treated with **selective estrogen receptor modulators**, we suggest treatment with low-dose vaginal estrogen or [vaginal DHEA](#) (prasterone) (**Grade 2C**). These medications are not associated with an increased risk of breast cancer recurrence. (See '[Patients with breast cancer](#)' above.)
  - For most patients treated with **aromatase inhibitors** (AIs), we suggest **not** using vaginal estrogen, DHEA (prasterone), or [testosterone](#) (**Grade 2C**). However, for those patients with marked symptoms of GSM who are willing to accept that vaginal hormonal therapy may limit the effectiveness of their cancer treatment, low-dose vaginal estrogen or prasterone is reasonable. Alternatively, some oncologists may switch treatment from AI therapy to [tamoxifen](#), depending on the risk features of the cancer and patient preferences, because of the lower risk of developing GSM. (See '[Patients with breast cancer](#)' above.)
- **Patients with refractory symptoms or severe anatomic changes** – Patients with refractory symptoms of GSM or severe anatomic changes (eg, introital narrowing, vaginal shortening) may benefit from pelvic physical therapy and vaginal dilators. Some patients with severe anatomic changes require surgery. Laser and other energy-based devices have been marketed for treatment of vulvovaginal atrophy (VVA), but the safety and efficacy of these devices remain uncertain. (See '[Symptoms refractory to hormone therapy](#)' above and '[Patients with severe anatomic changes](#)' above and '[Laser or radiofrequency devices](#)' above.)

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## GRAPHICS

Table 1: Commonly available nonhormonal vaginal moisturizers formulations

| Product                                  | Ingredients                                                                                                                                                                                                                                                    | pH <sup>[1,2]</sup> | Osmolality (mOSm/kg) <sup>[1,2]</sup> | Formulation and use                                                 | Approximate price                      |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------|---------------------------------------------------------------------|----------------------------------------|
| RepHresh Vaginal gel                     | Purified water USP, glycerin, polycarbophil, carbopol 974P, ethylparaben sodium, methylparaben sodium, propylparaben sodium, sodium hydroxide                                                                                                                  | 3.4                 | 1439 to 1914                          | Prefilled applicator used every 3 days                              | 4 applicators (12 days) \$26.49        |
| Replens Long-Lasting Vaginal Moisturizer | Purified water, glycerin, mineral oil, polycarbophil, carbomer homopolymer type B, hydrogenated palm oil glyceride, sorbic acid, sodium hydroxide                                                                                                              | 2.95 to 3.0         | 1177 to 2011                          | Prefilled applicator used every 3 days                              | 8 applicators (1-month supply) \$22.49 |
| Luvena                                   | Water, propanediol, PEG-20, xanthan gum, PEG-20M, simmonsia chinensis (Jojoba) seed oil, vaccinium macrocopon (cranberry) fruit extract, lysozyme, lactoferrin, lactoperoxidase, lactic acid, potassium thiocyanate, glycogen, mannose, tocopherol (vitamin E) | Unavailable         | Unavailable                           | Prefilled applicators, use every 3 days as moisturizer or lubricant | 6 applicators for \$19.99              |
| Canesintima Intimate                     | Aqua, glycerin, glyceryl                                                                                                                                                                                                                                       | 5.63                | 846                                   | Pump bottle                                                         | 50 mL pump bottle for                  |

|                                        |                                                                                                                                                                                                                                                                             |            |              |                                                                   |                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Moisturizer                            | polymethacrylate, capryloyl glycine, sorbitol, acrylates/C10-30 alkyl acrylate crosspolymer, sodium hyaluronate, sodium benzoate, sodium hydroxide, galactoarabinan, butylene glycol/camellia japonica leaf/flower extract, tetrasodium EDTA, P-anisic acid, levulinic acid |            |              |                                                                   | \$33.49                                                 |
| Ah! Yes Vaginal Moisturizer            | Aqua (water), linum usitatissimum (flax) extract, aloe barbadensis leaf juice powder (aloe vera), ceratonia siliqua (locust bean gum), cyamopsis tetragonolobus (guar gum), sodium chloride, xanthan gum, potassium sorbate, citric acid, phenoxyethanol                    | 3.8 to 4.2 | 260 to 290   | Flip top bottle (100 mL); prefilled applicators used every 3 days | \$14.99 for flip top bottle, \$53.89 for 30 applicators |
| Sylk Natural Intimate Moisturizer      | Purified water, kiwifruit plant extract, citrus seed extract, xanthan plant extract, vegetable glycerine, citric acid, potassium sorbate, sodium citrate                                                                                                                    | 4.47       | 877          | 3 fluid oz flip top tube                                          | \$22.95                                                 |
| HyaloSyl Natural Vaginal Hydrating Gel | Hydeal-D® (hyaluronic acid derivative), propylene glycol, carbomer (carbopol 974P), methyl p-hydroxybenzoate, propyl p-hydroxybenzoate,                                                                                                                                     | 4.88       | 1336 to 1729 | 30 g tube with 10 applicators (30-day supply)                     | \$27.00                                                 |

|                                             |                                                                                                                                                                                                                                   |             |             |                                                |         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|------------------------------------------------|---------|
|                                             | sodium hydroxide,<br>purified water                                                                                                                                                                                               |             |             |                                                |         |
| Revaree<br>Vaginal<br>Moisturizer           | 2-g vaginal insert<br>contains 5 mg of<br>hyaluronic acid<br>sodium salt in a base<br>consisting of a<br>mixture of semi-<br>synthetic glycerides,<br>which help it retain<br>its shape. (Full<br>ingredient list not<br>listed.) | Unavailable | Unavailable | 10 vaginal<br>suppositories<br>(30-day supply) | \$55.00 |
| K-Y<br>Liquibeads<br>Vaginal<br>Moisturizer | Dimethicone,<br>dimethiconol,<br>gelatin, glycerin                                                                                                                                                                                | Unavailable | Unavailable | 6 vaginal<br>ovules                            | \$22.79 |

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<https://www.mdedge.com/obgyn> (Accessed on August 9, 2022).

Table 2: Commonly available nonhormonal lubricants

| Product                                                 | Ingredients                                                                                                                                                      | pH <sup>[1-3]</sup> | Osmolality (mOSm/kg) <sup>[1-3]</sup> | Formulation and use        | Approximate price  |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------|----------------------------|--------------------|
| K-Y Jelly Water Based Lubricant                         | Water, glycerin, hydroxyethylcellulose, chlorhexidine, gluconate, gluconolactone, methylparaben, sodium hydroxide                                                | 3.5 to 4.49         | 2007 to 3631                          | 2 and 4 oz flip top bottle | \$4.99 to \$7.59   |
| Astroglide Gel Lubricant                                | Purified water, glycerin, hydroxyethylcellulose, chlorhexidine gluconate, methylparaben, glucono delta lactone, sodium hydroxide                                 | 4.38                | 6100                                  | 4 oz flip top tube         | \$6.79             |
| ID Glide Lubricant                                      | Water, glycerin, propylene glycol, cellulose gum, EDTA, carbomer, PEG-90M, tetrahydroxypropyl ethylenediamine, methylparaben, sodium benzoate, potassium sorbate | 5.20*               | 3200*                                 | Multiple                   | \$3.49 to \$159.99 |
| Wet Original Gel Lubricant                              | Water (aqua), glycerin, carboxymethylcellulose, pentylene glycol, potassium sorbate                                                                              | 5.9                 | 3679                                  | 5 to 32 oz bottles         | \$8.33 to \$39.51  |
| Durex Play Feel Pleasure Gel                            | Purified water, propylene glycol, hydroxyethylcellulose, benzoic acid, sodium hydroxide                                                                          | 5.48                | 1563                                  | 50 mL flip top bottle      | \$8.99             |
| Good Clean Love Almost Naked Organic Personal Lubricant | Organic aloe barbadensis leaf juice, xanthan gum, agar, lactic acid, potassium sorbate, sodium benzoate, natural flavor                                          | 4.73                | 240                                   | 4 oz flip top tube         | \$11.99            |
| H2O Sliquid Naturals                                    | Purified water, plant cellulose (from cotton), cyamopsis (guar                                                                                                   | 4.0 to 4.4          | Unavailable                           | 4.2 or 8.5 oz bottles      | \$12.00 to \$19.00 |

|                                           |                                                                                                                                                                                                                                                               |      |       |                                 |                    |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------------------------------|--------------------|
|                                           | conditioners), potassium sorbate, citric acid                                                                                                                                                                                                                 |      |       |                                 |                    |
| Pjur Med Natural Glide Personal Lubricant | Aqua (water), glycerin, xanthan gum, benzyl alcohol, sodium benzoate, potassium sorbate, citric acid                                                                                                                                                          | 4.41 | >2000 | 100 mL bottle                   | \$15.95            |
| Pjur Women Nude Lubricant                 | Aqua (water), propylene glycol, ethoxydiglycol, hydroxypropyl guar hydroxypropyltrimonium chloride, hydroxyethyl cellulose, sodium saccharin, citric acid                                                                                                     | 4.42 | >2000 | 100 mL bottle                   | \$34.99            |
| Ritex Sensitiv Gel                        | Aqua, glycerin, butylene glycol, hydroxyethylcellulose, sodium lactate, lactic acid                                                                                                                                                                           | 4.04 | >2000 | 100 mL tube                     | \$48.99            |
| Simply Slick Personal Lubricating Lotion  | Castor oil, purified water, jojoba oil, vegetable glycerin, pectin, Stevia, Optiflo H370VF                                                                                                                                                                    | 6.68 | >2000 | 2 oz bottle                     | \$17.77            |
| Ah! Yes Water Based Lubricant             | Aqua (water), aloe barbadensis leaf juice powder (aloe vera), linum usitatissimum (flax) seed extract, cyamopsis tetragonolobus (guar gum), ceratonia siliqua (locust bean gum), xanthan gum, sodium chloride, potassium sorbate, citric acid, phenoxyethanol | 4.08 | 154   | 1.7 to 5.1 oz flip top tubes    | \$8.49 to \$21.99  |
| Uber Lube                                 | Dimethicone, dimethiconol, cyclomethicone, tocopheryl acetate (vitamin E)                                                                                                                                                                                     | 4.6  | N/A   | 50 and 100 mL pump glass bottle | \$18.00 to \$28.00 |
| Swiss Navy Silicone Lubricant             | Cyclopentasiloxane, dimethicone, tocopheryl acetate                                                                                                                                                                                                           | N/A  | N/A   | 4 oz pump bottle                | \$16.49            |
| K-Y True Feel Premium                     | Dimethicone                                                                                                                                                                                                                                                   | N/A  | N/A   | 1.5 oz bottle                   | \$13.99            |

\* Formulation of product changed since laboratory testing so pH and osmolality may be different.

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**Table 3: Vaginal hormone preparations available in the United States for treatment of genitourinary syndrome of menopause (vulvovaginal atrophy)**

| Preparation             |                            | United States trade name | Available strength(s)                            | Suggested regimen                                                                                                                                                                                                                                                      |
|-------------------------|----------------------------|--------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Estrogens</b>        |                            |                          |                                                  |                                                                                                                                                                                                                                                                        |
| Conjugated estrogen     | Vaginal cream <sup>*</sup> | Premarin                 | 0.625 mg conjugated estrogens per g of cream     | 0.5 g of cream intravaginally administered once daily for 2 weeks, then reduce to twice weekly <sup>¶</sup> ; dose may be adjusted according to patient's symptoms and clinical response (range: 0.5 to 2 g of cream).                                                 |
| Estradiol               | Vaginal cream <sup>*</sup> | Estrace                  | 100 mcg estradiol per g of cream                 | 0.5 g of cream intravaginally administered once daily for 2 weeks, then reduce to twice weekly <sup>¶</sup> ; dose may be adjusted according to patient's symptoms and clinical response (initial range: 0.5 to 4 g of cream; maintenance range: 0.5 to 1 g of cream). |
|                         | Vaginal insert             | Imvexxy                  | 4 mcg or 10 mcg estradiol per vaginal insert     | Insert 1 softgel intravaginally once daily for 2 weeks, followed by twice weekly. Initial dose depends on clinical severity and patient preference; dose may be increased or decreased according to clinical response.                                                 |
|                         |                            | Vagifem, Yuvaferm        | 10 mcg estradiol per vaginal insert              | Insert 1 tablet intravaginally once daily for 2 weeks, followed by twice weekly.                                                                                                                                                                                       |
|                         | Vaginal ring               | Estring                  | 7.5 mcg estradiol per day, released over 90 days | Ring is inserted into the vagina by the patient or clinician. Ring is removed and replaced with a new ring every 90 days.                                                                                                                                              |
| <b>DHEA<sup>A</sup></b> |                            |                          |                                                  |                                                                                                                                                                                                                                                                        |

|            |                |           |                                      |                                                 |
|------------|----------------|-----------|--------------------------------------|-------------------------------------------------|
| Prasterone | Vaginal insert | Intrarosa | 6.5 mg prasterone per vaginal insert | Insert 1 suppository intravaginally once daily. |
|------------|----------------|-----------|--------------------------------------|-------------------------------------------------|

For clinical indications, refer to related UpToDate content. Product choice is often determined by patient preference and cost.

DHEA: dehydroepiandrosterone; GSM: genitourinary syndrome of menopause.

\* Vaginal estrogen creams are supplied with an applicator marked with dose calibrations so the patient can measure the prescribed amount of cream.

¶ Our practice differs from the manufacturer's product information; refer to UpToDate topic for further information.

Δ While approved only for the indication of GSM-associated dyspareunia, studies have also demonstrated improvement in vaginal dryness. This is discussed in related UpToDate content.

*Prepared with data from:*

1. FDA prescribing information available at US National Library of Medicine NIH DailyMed website. Available at: <https://dailymed.nlm.nih.gov/dailymed/index.cfm> (Accessed on October 4, 2024).
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## Contributor Disclosures

**Gloria Bachmann, MD, MMS** No relevant financial relationship(s) with ineligible companies to disclose. **JoAnn V Pinkerton, MD, FACOG, MSCP** Grant/Research/Clinical Trial Support: Bayer [Nonhormone therapy for vasomotor symptoms]. Consultant/Advisory Boards: Bayer Pharmaceuticals [Nonhormone therapy for vasomotor symptoms]; Pfizer [Review of evidence for removing boxed warning for low-dose vaginal estrogens [did not discuss products]]. All of the relevant financial relationships listed have been mitigated. **Robert L Barbieri, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Harold J Burstein, MD, PhD** No relevant financial relationship(s) with ineligible companies to disclose. **Alana Chakrabarti, MD, FACOG** No relevant financial relationship(s) with ineligible companies to disclose.

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